

3	(a) resultant force (in any direction) is zero resultant moment/torque (about any point) is zero	B1	B1	[2]
	(b) (i) force = $33 \sin 52^\circ$ or $33 \cos 38^\circ$ = 26 N	A1	[1]	
	(ii) 26×0.30 or $W \times 0.20$ or 12×0.40	C1		
	$26 \times 0.30 = (W \times 0.20) + (12 \times 0.40)$	C1		
	$W = 15 \text{ N}$	A1	[3]	
	(c) (i) $E = \Delta\sigma / \Delta\varepsilon$ or $E = \sigma / \varepsilon$	C1		
	$\Delta\sigma = 2.0 \times 10^{11} \times 7.5 \times 10^{-4}$ = $1.5 \times 10^8 \text{ Pa}$	A1	[2]	
	(ii) $\Delta\sigma = \Delta F / A$ or $\sigma = F / A$	C1		
	$A = 78 / 1.5 \times 10^8$ ($= 5.2 \times 10^{-7} \text{ m}^2$)	C1		
	$5.2 \times 10^{-7} = \pi d^2 / 4$			
	$d = 8.1 \times 10^{-4} \text{ m}$	A1	[3]	